

Octubre 1

2018

1) Sea V el subespacio en $\mathbb{R}^4$ que consiste de todas las soluciones del siguiente sistema de ecuaciones lineales homogéneas

$$2x - y + 2z + w = 0$$

$$x + y + z - w = 0$$

$$2x + 4z - w = 0$$

Determina una base para V

$$\left(\begin{array}{cccc|c} 2 & -1 & 2 & 1 & 0 \\ 1 & 1 & 1 & -1 & 0 \\ 2 & 0 & 4 & -1 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 2 & -1 & 2 & 1 & 0 \\ 2 & 0 & 4 & -1 & 0 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & -3 & 0 & 3 & 0 \\ 0 & -2 & 2 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & 2 & 1 & 0 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & -2 & 2 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 2 & -1 & 0 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1/2 & 0 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 1 & 0 & -1/2 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1/2 & 0 \end{array} \right)$$

$$x + y = \frac{1}{2}w$$

$$y = w$$

$$2z = w$$

$$2x + 2y = w$$

$$y = w$$

$$2z = w$$

$$2x = -w$$

$$y = w$$

$$2z = w$$

$$x = -\frac{w}{2}$$

$$y = w$$

$$z = \frac{w}{2}$$

$\Rightarrow$

$$w \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \\ 1 \end{pmatrix} \text{ así } V = \left\{ \text{gen} \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \\ 1 \end{pmatrix} \right\}$$

Video 26
Clase 14

2) Mostrar que si $a \neq b$, entonces

$$\det \begin{pmatrix} a+b & ab & 0 & \dots & 0 & 0 \\ 1 & a+b & ab & \dots & 0 & 0 \\ 0 & 1 & a+b & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & a+b & ab \\ 0 & 0 & 0 & \dots & 1 & a+b \end{pmatrix} = \frac{a^{n+1} - b^{n+1}}{a-b}$$

donde la matriz es nxn

Notemos

$$\begin{array}{r} a^n + a^{n-1}b + a^{n-2}b^2 + \dots + ab^{n-1} + b^n \\ a-b \mid a^{n+1} - b^{n+1} \\ \underline{-a^{n+1} + a^n b} \\ a^n b - b^{n+1} \\ \underline{-a^n b + a^{n-1} b^2} \\ a^{n-1} b^2 - b^{n+1} \\ \underline{-a^{n-1} b^2 + a^{n-2} b^3} \\ a^{n-2} b^3 - b^{n+1} \\ \vdots \\ a^2 b^{n-1} - b^{n+1} \\ \underline{-a^2 b^{n-1} + ab^n} \\ ab^n - b^{n+1} \\ \underline{-ab^n + b^{n+1}} \\ 0 \end{array}$$

Entonces

$$\frac{a^{n+1} - b^{n+1}}{a-b} = a^n + a^{n-1}b + a^{n-2}b^2 + \dots + ab^{n-1} + b^n$$

Veamos los casos 2x2

$$\begin{vmatrix} a+b & ab \\ 1 & a+b \end{vmatrix} = (a+b)^2 - ab = a^2 + ab + b^2$$

Caso 3x3

$$\begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} = (a+b) \begin{vmatrix} a+b & ab \\ 1 & a+b \end{vmatrix} - ab \begin{vmatrix} 1 & ab \\ 0 & a+b \end{vmatrix}$$

$$= (a+b) [a^2 + ab + b^2] - ab(a+b) = (a+b)a^2 + (a+b)ab + (a+b)b^2$$

$$- ab(a+b) = (a+b)a^2 + (a+b)b^2 = a^3 + a^2b + ab^2 + b^3$$

Caso 4x4

$$\begin{vmatrix} a+b & ab & 0 & 0 \\ 1 & a+b & ab & 0 \\ 0 & 1 & a+b & ab \\ 0 & 0 & 1 & a+b \end{vmatrix} = (a+b) \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix} - ab \begin{vmatrix} 1 & ab & 0 \\ 0 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix}$$

$$(a+b) [a^3 + a^2b + ab^2 + b^3] - ab \begin{vmatrix} a+b & ab \\ 1 & a+b \end{vmatrix} =$$

$$(a+b)a^3 + (a+b)a^2b + (a+b)ab^2 + (a+b)b^3 - ab[(a+b)^2 - ab] =$$

$$a^4 + a^3b + a^3b + a^2b^2 + a^2b^2 + ab^3 + ab^3 + b^4 - a^3b - 2a^2b^2 - ab^3 + a^2b^2 =$$

$$a^4 + a^3b + a^2b^2 + ab^3 + \underline{ab^4}$$

Caso 5x5

$$\begin{vmatrix} a+b & ab & 0 & 0 & 0 \\ 1 & a+b & ab & 0 & 0 \\ 0 & 1 & a+b & ab & 0 \\ 0 & 0 & 1 & a+b & ab \\ 0 & 0 & 0 & 1 & a+b \end{vmatrix} = (a+b) \begin{vmatrix} a+b & ab & 0 & 0 \\ 1 & a+b & ab & 0 \\ 0 & 1 & a+b & ab \\ 0 & 0 & 1 & a+b \end{vmatrix}$$

$$- ab \begin{vmatrix} a+b & ab & 0 \\ 1 & a+b & ab \\ 0 & 1 & a+b \end{vmatrix}$$

Matriz 3x3

Así para el caso $n \times n$ sea

← Determinante de matriz $(n-1) \times (n-1)$

$$(a+b) \cdot |A_{n-1}| - ab |A_{n-2}| \leftarrow \text{Determinante matriz } (n-2) \times (n-2)$$

$$= (a+b) [a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1}] -$$

$$ab [a^{n-2} + a^{n-3}b + \dots + ab^{n-3} + b^{n-2}]$$

$$= a^n + a^{n-1}b + a^{n-2}b^2 + \dots + a^2b^{n-2} + ab^{n-1} + a^{n-1}b + a^{n-2}b^2 + \dots + ab^{n-1} + b^n$$
$$- [a^{n-1}b + a^{n-2}b^2 + \dots + ab^{n-1}]$$

$$= a^n + 2a^{n-1}b + 2a^{n-2}b^2 + \dots + 2a^2b^{n-2} + 2ab^{n-1} + b^n$$
$$- [a^{n-1}b + a^{n-2}b^2 + \dots + ab^{n-1}]$$

$$= a^n + a^{n-1}b + a^{n-2}b^2 + a^{n-3}b^3 + \dots + a^2b^{n-2} + ab^{n-1} + b^n$$

$$= \frac{a^{n+1} - b^{n+1}}{a - b}$$

Q.E.D.

$$\begin{array}{cccc|cccc} -\lambda & -\lambda^2 & 0 & 0 & -\lambda & 0 & 0 & 0 \\ \lambda & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ \hline 0 & 1-\lambda^2 & 0 & 0 & -\lambda & 1 & 0 & 0 \end{array}$$

3) ¿Para que valores de λ es invertible la matriz?

$$A = \begin{pmatrix} 1 & \lambda & 0 & 0 \\ \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \end{pmatrix} \quad \frac{1}{\det A} A^+ = A^{-1}$$

$$\left(\begin{array}{cccc|cccc} 1 & \lambda & 0 & 0 & 1 & 0 & 0 & 0 \\ \lambda & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 1 \end{array} \right) F_2 - \lambda F_1$$

$$\left(\begin{array}{cccc|cccc} 1 & \lambda & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1-\lambda^2 & 0 & 0 & -\lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 1 \end{array} \right) F_2 \left(\frac{1}{1-\lambda^2} \right)$$

$$\left(\begin{array}{cccc|cccc} 1 & \lambda & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{\lambda}{1-\lambda^2} & \frac{1}{1-\lambda^2} & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{cccc|cccc} 0 & -\lambda & 0 & 0 & +\frac{\lambda^2}{1-\lambda^2} & -\frac{\lambda}{1-\lambda^2} & 0 & 0 \\ 1 & \lambda & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & \frac{1}{1-\lambda^2} & -\frac{1}{1-\lambda^2} & 0 & 0 \end{array}$$

$$\begin{array}{cccc|cccc} 0 & -\lambda & 0 & 0 & \frac{\lambda^2}{1-\lambda^2} & -\frac{1}{1-\lambda^2} & 0 & 0 \\ 0 & \lambda & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & \frac{\lambda^2}{1-\lambda^2} & -\frac{1}{1-\lambda^2} & 1 & 0 \end{array}$$

$$\left(\begin{array}{cccc|cc} 1 & 0 & 0 & 0 & 1-\lambda^2 & 1-\lambda^2 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{\lambda}{1-\lambda^2} & \frac{1}{1-\lambda^2} & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{\lambda^2}{1-\lambda^2} & -\frac{1}{1-\lambda^2} & 1 & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$\begin{array}{cccc|cc} 0 & 0 & -\lambda & 0 & -\frac{\lambda^3}{1-\lambda^2} & +\frac{\lambda}{1-\lambda^2} & -\lambda & 0 \\ 0 & 0 & \lambda & 1 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 1 & -\frac{\lambda^3}{1-\lambda^2} & \frac{\lambda}{1-\lambda^2} & -\lambda & 1 \end{array}$$

$$\left(\begin{array}{cccc} 1 & \lambda & 0 & 0 \\ \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \end{array} \right) = - \left| \begin{array}{ccc} 1 & \lambda & 0 \\ \lambda & 1 & 0 \\ 0 & \lambda & 1 \end{array} \right| = - \left| \begin{array}{cc} 1 & \lambda \\ \lambda & 1 \end{array} \right|$$

4) Sea $T(x, y, z, w) = (x - y + 2z + 3w, y + 4z + 3w, x + 6z + 6w)$

a) ~~Determine~~ Encuentre la representación matricial A de T con respecto a la base canónica

$$T(1, 0, 0, 0) = (1, 0, 1)$$

$$T(0, 1, 0, 0) = (-1, 1, 0)$$

$$T(0, 0, 1, 0) = (2, 4, 6)$$

$$T(0, 0, 0, 1) = (3, 3, 6)$$

Así la matriz es

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 4 & 3 \\ 1 & 0 & 6 & 6 \end{pmatrix}$$

b) Determine los subespacios núcleo e imagen de T y encuentre sus dimensiones.

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 4 & 3 \\ 1 & 0 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 4 & 3 \\ 0 & 1 & 4 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 6 & 6 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{Im } T = \left\{ \text{gen} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\} \right\}$$

$$\begin{aligned} x + 6z + 6w &= 0 \Rightarrow x = -6z - 6w \\ y + 4z + 3w &= 0 \Rightarrow y = -4z - 3w \end{aligned} \quad \text{Nul } T = \left\{ \text{gen} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\} \right\}$$

$$\begin{pmatrix} -6z - 6w \\ -4z - 3w \\ z \\ w \end{pmatrix} = z \begin{pmatrix} -6 \\ -4 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} -6 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{-1 + \sqrt{5}}{2} \\ 1 \end{pmatrix}$$

5) Dada la transformación lineal $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$,

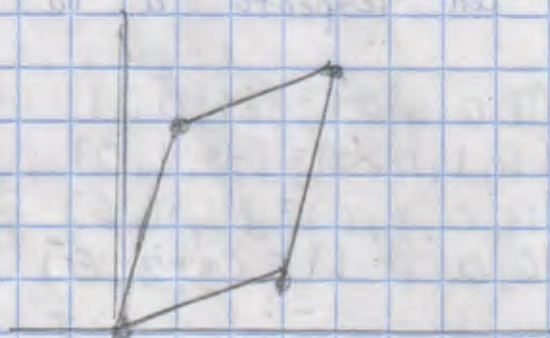
$$T(x, y) = (3x + y, x + 4y)$$

$$T(0, 0) = (0, 0)$$

$$T(1, 0) = (3, 1)$$

$$T(1, 1) = (4, 5)$$

$$T(0, 1) = (1, 4)$$



$$\begin{pmatrix} \frac{-\sqrt{5}-1}{2} \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}$$

$$\lambda x = 3x + y$$

$$\lambda y = x + 4y$$

$$(\lambda - 3)x = y$$

$$(\lambda - 4)y = x$$

$$\lambda x = \frac{y}{\lambda - 3}$$

$$x = (\lambda - 4)y$$

$$x + 4y = y$$

$$(\lambda - 4)y = \frac{y}{\lambda - 3}$$

$$\begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}$$

$$(3 - \lambda)(4 - \lambda) - 1$$

$$(\lambda - 4)(\lambda - 3)y - y = 0$$

$$y(\lambda - 4)(\lambda - 3) - 1 = 0$$

$$\begin{pmatrix} \frac{-1 + 3\sqrt{5}}{2} \\ \frac{7 + \sqrt{5}}{2} \end{pmatrix}$$